

Environmental recurrence selects a costly memory-use allele without a resolved one-bit population benefit

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Abstract

Prior theory and experiments establish that environmental predictability can favor memory-like inheritance, learning and anticipatory effects, and that memory can carry costs. We asked a narrower, prospectively specified question in an abstract computational model. Does lag-two environmental recurrence enrich a mutable allele for using a supplied one-record private cache when each retrieval displaces one of two fresh exploratory proposals, and does any such enrichment coincide with a meaningful gain in population accuracy? In a finite population of 32 haploid searchers, a binary use policy mutated symmetrically at rate $1/32$ and was selected only through the objective scores of the candidates it caused to be proposed. A label-blind SHAM arm supplied an exact neutral benchmark for the label process. Across 41,600 independent paired blocks (332,800 paths and 10,905,190,400 objective queries), late-window start contrasts were bounded within one expected individual (95% familywise) over the 256-update horizon, passing the prespecified start-state gate, and recurrence strongly enriched the memory-use allele. Relative to the neutral $1/2$, the start-averaged excess was 0.310292 (95% familywise Hoeffding-Bonferroni interval [0.302482, 0.318102]); relative to the nonrecurrent law it was 0.441635 ([0.426015, 0.457256]). By contrast, the recurrent-law ACTIVE minus SHAM accuracy contrast was 0.009377 ([-0.005138, 0.023892]), bounded below the prespecified one-correct-bit scale of 0.03125, with sign unresolved, and the recurrence interaction in accuracy, 0.011360 ([-0.017669, 0.040390]), was unresolved. Under the nonrecurrent law, ACTIVE accuracy was descriptively 0.001983 below SHAM. Under one deliberately aligned recurrence law and one fixed opportunity cost, selection on an information-use policy therefore separated from consequential population performance. The result does not address the origin of memory architecture, other recurrence laws, other costs or biological systems.

1. Introduction

Recurring environments make the past informative about the future, and a substantial literature shows that this can favor carrying information across time. Kuijper and Johnstone found that environmental stability and predictability can favor positive parental effects and long-term phenotypic memory under genetically determined transmission rules [1]. Carja and Plotkin showed that, in periodically fluctuating environments, an intermediate degree of phenotypic memory maximizes the fixation probability of an allele that permits heritable phenotypic heterogeneity [2]. Dey, Proulx and Teotónio showed experimentally that temporal environmental structure can select maternal effects [8]; Skanata and Kussell derived evolutionary regimes for response networks with phenotypic memory in random environments [9]; and Kronholm found that anticipatory effects mediated by epigenetic changes evolve when an environmental cue is reliable, the environment changes at intermediate rates and the costs of the effect are low [7].

Models of learning reach related conclusions. Wright and colleagues allowed memory weighting and current sampling effort to evolve under explicit costs and found that memory increased with temporal autocorrelation and interacted with costly sampling [3]. Arehart and Adler quantified the costs and benefits of memory duration and exploration in a minimal learning model and found that the marginal benefit of learning is often small [4]. Morgan, Suchow and Griffiths inserted human participants as agents into evolutionary simulations with simulated loci for learning and memory and found that rapid environmental change dampened their evolution [5]. Among the studies identified by our bounded search, this is a close precedent for an evolving memory-use trait, although its learners were human participants rather than a cleanly mutable allele. Other work is at least as close on other dimensions: Carja and Plotkin analyzed a mutable modifier allele and its fixation in a finite population [2], and Grabowski and colleagues studied the early evolution of memory usage in digital organisms [6].

The value and cost of memory that is already present are also well studied. Dunlap and Stephens linked environmental change to optimal sampling and memory weighting [10]. Lambert and Kussell measured bacterial memory under fluctuating carbon sources and modeled its long-term fitness benefit [11]. Rescan and colleagues linked phenotypic memory to population growth and extinction risk in a noisy environment [12]. George optimized memory-driven phenotypic adaptation in fluctuating environments [13], and Jain, Jolly and George added adaptation costs [14]. Pollack, Nozoe and Kussell analyzed selection on gene regulation and phenotypic memory under metabolic fluctuations [15]. Leadbeater and Watrobska emphasize that memory can be shaped by selection along many mechanistic dimensions rather than as a single scalar trait [16].

Against this background, we do not claim to show that memory evolves, that predictability favors memory, or that costly memory can be selected; those propositions are occupied by the work above. The contribution here is a specific joint design, executed with a new implementation and a new random cohort, that allows two outcomes to be judged separately. First, the memory architecture, a one-record private cache, is supplied, and only a binary use policy mutates and is selected, in a finite population with explicitly stochastic survival. Second, the cost of using memory is a one-for-one search-budget opportunity cost: a carrier that retrieves its cache gives up one of its two fresh proposals, with no added fitness penalty or tunable coefficient. Third, a label-blind SHAM arm supplies an exact neutral benchmark for the label process at the same query budget. Fourth, allele enrichment and population accuracy were judged by prospectively separate decision rules, each with a stated meaningful scale.

The separation matters because the two outcomes need not move together. An allele can rise in frequency because it improves its carriers' chances of being among the survivors while the population's average fit to the environment changes little. Carriers may displace non-carriers that would have achieved similar accuracy, or gains at recurrent updates may be offset by lost exploration at non-recurrent updates. Treating allele enrichment as evidence of population benefit would conflate these possibilities.

We report the accepted result of the prospective study PHASE2-MUTABLE-MEMORY-001. Under one recurrent target law that was deliberately aligned with the cache delay, lag-two recurrence strongly enriched the mutation-generated memory-use allele above both its exact neutral expectation and its frequency under a nonrecurrent control, despite the fixed opportunity cost. The corresponding ACTIVE minus SHAM accuracy contrast was bounded below the prespecified one-correct-bit scale, with sign unresolved, and the recurrence interaction in accuracy remained unresolved. We present these two findings as a single result: in this model, recurrence selected a costly memory-use policy without a demonstrated commensurate population-performance advantage at the fixed scale.

2. Model and prospective design

The specification, decision table, cohort size, audit plan and claim limits were frozen before any production outcome was generated. A non-implementing AI reviewer returned a terminal GO review of the frozen specification before production was authorized; the specification hash, review record and timestamps are given in Section 7. No parameter, cohort, horizon, target law, mutation rate or cost was changed after outcomes were observed. The complete update order is given in Supplementary Sections S2-S7.

2.1 Population, search state and objective

The population contains 32 haploid individuals in fixed slots 0-31. Each individual carries an unsigned 32-bit genotype and a binary policy allele, M (use the private cache) or F (fresh exploration). An M individual also holds a private cache that is either invalid or contains one 32-bit genotype; F individuals hold no valid cache. The allele jointly determines whether the one-step record is retained and whether it is retrieved; every individual has a cache slot, but only M carriers receive a valid record, so this design cannot separate selection on retention from selection on retrieval. Lineage identifiers are audit metadata only and are never read by any scientific operator. Each path runs for updates 1-256. At update t the environment presents a 32-bit target T_t , and the raw objective of a genotype is its Hamming mismatch to T_t . Population accuracy at an update is one minus the mean survivor mismatch divided by 32, measured after survival; a population of uniformly random genotypes has expected accuracy $1/2$. The late window is updates 193-256 inclusive.

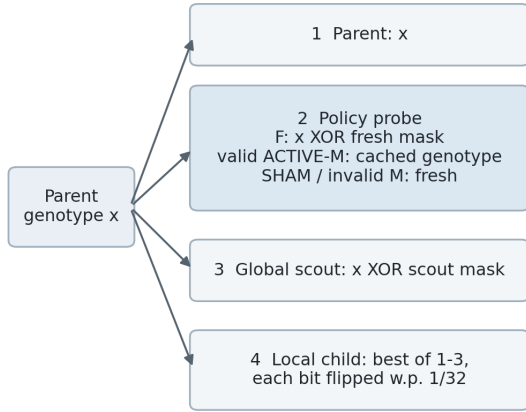
The population size, bit length, four-candidate architecture, lag-two law and selection weights were inherited from earlier internal private-memory studies so that the changed question, selection on a mutable use policy, would be identifiable. The implementation and all random cohorts are new and import no producer code from those studies. This study is not a replication of them.

2.2 Candidate budget and the one-proposal opportunity cost

Every parent produces exactly four evaluated candidates per update, so every update makes exactly 128 objective queries regardless of allele, arm or law. The candidates are (1) the unchanged parent; (2) a policy probe; (3) a global scout, the parent XOR an independent uniform 32-bit mask; and (4) a local child, made by taking the lowest-mismatch member of candidates 1-3 (ties broken by a keyed random key and then by family order) and flipping each of its 32 bits independently with probability $1/32$. For an F individual the policy probe is the parent XOR a second independent uniform mask, so an F carrier makes two fresh, effectively uniform proposals per update. For a valid M carrier in the ACTIVE arm, the policy probe is the cached genotype itself. Because the probe is also eligible to be the donor of the local child, a useful cache can shape two of the four candidates.

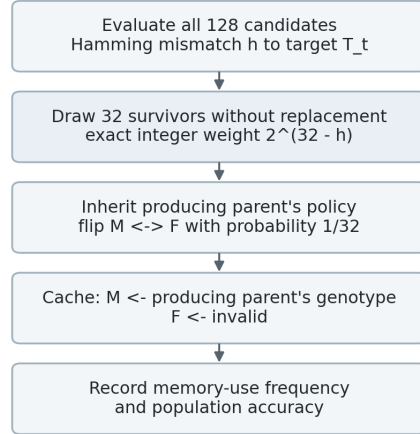
The cost of memory is therefore an opportunity cost on a fixed search budget: retrieval replaces exactly one of two effectively uniform fresh proposals with the cached record. There is no score penalty and no tunable cost coefficient. This is a model-specific search-budget trade-off, not a calibrated energetic, metabolic or biological cost, and its magnitude is conditional on the frozen fresh-probe distribution. When the cache is uninformative, an M carrier loses one fresh draw at the same query count. When the cache closely matches the current target, retrieval supplies a candidate that a uniform fresh draw is unlikely to match. Genotypes are not deduplicated: if the cache equals the unchanged parent, both copies are separately evaluated and the update still makes 128 queries.

(a) Four candidates per parent per update



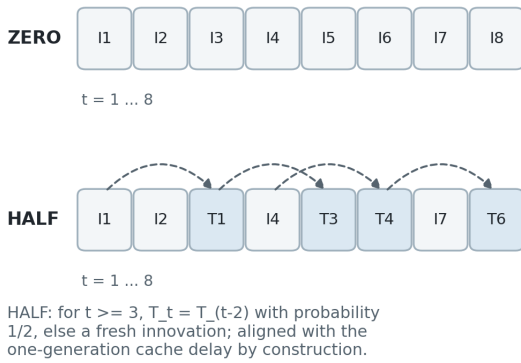
Retrieval replaces one of two uniform fresh proposals; 128 objective queries per update in every cell.

(b) Population event order at update t



No label, cache or lineage identifier enters selection.

(c) Target laws (illustrative, not data)



(d) Paired cells and inferential families

Each block: 2 arms (ACTIVE, SHAM) x 2 laws (ZERO, HALF) x 2 starts (ALL_F, ALL_M) = 8 paired cells sharing all keyed random values. SHAM never reads labels or caches: exact neutral label benchmark.

Primary family ($\alpha_{\text{each}} = 0.05/4$): allele
 $C_{\text{abs}} = A_{\text{HALF}} - 1/2$; $C_{\text{rec}} = A_{\text{HALF}} - A_{\text{ZERO}}$
Start-state gate: $D_{\text{HALF}}, D_{\text{ZERO}}$
inside $(-1/32, +1/32)$

Secondary family ($\alpha_{\text{each}} = 0.05/2$): accuracy
 $P_{\text{abs}} (\text{HALF: ACTIVE} - \text{SHAM})$; $P_{\text{rec}} (\text{interaction})$
Scale: one correct bit per survivor $(1/32)$;
cannot alter the primary decision

Figure 1. Model and prospective design. (a) One update for one parent: four evaluated candidates, of which only the policy probe depends on the allele and arm; a valid ACTIVE-M carrier replaces one of its two fresh uniform proposals with its cached genotype. (b) Population-level event order: evaluation of 128 candidates against the current target, exact integer-weighted sequential sampling of 32 survivors, symmetric policy mutation, and cache assignment. (c) The two target laws; HALF returns to the target from two updates earlier with probability $1/2$, which aligns with the one-generation cache delay by construction. (d) The eight paired cells per block and the separate primary (allele) and secondary (accuracy) inferential families. The schematic contains no outcome values.

2.3 Target laws and the deliberate HALF alignment

For every block and update, an innovation I_t (a uniform 32-bit word) and a fair recurrence bit R_t are generated whether or not they are used. Under ZERO, $T_t = I_t$ at every update. Under HALF, $T_1 = I_1$ and $T_2 = I_2$; for $t \geq 3$ the target returns to $T_{(t-2)}$ when $R_t = 1$ and otherwise equals I_t . Non-recurrent HALF updates use the same innovation as ZERO at that update.

The cache of a survivor records its producing parent's genotype from the update in which the survivor was made, and that parent had itself survived selection against the previous target. A cache used at update $t+1$ therefore holds a genotype that survived selection against $T_{(t-1)}$, which under HALF is the target at update $t+1$ with probability $1/2$. HALF was a prior, outcome-informed choice made precisely to align the one-generation cache delay with a two-update target return; it favors retrieval by construction. No other lag, copy probability, target

law, cost or model class was tested.

2.4 Selection, policy mutation and cache inheritance

All 128 candidates are evaluated. Candidate j with mismatch h_j receives the exact integer weight $2^{(32-h_j)}$, proportional to $2^{(-h_j)}$. Thirty-two survivors are drawn sequentially without replacement, each with probability proportional to its weight among the candidates not yet selected. This is an exact Plackett-Luce weighted sample implemented entirely in unsigned integer arithmetic, with unbiased bounded integers produced by multiply-high rejection (Supplementary Section S5). No label, cache, lineage identifier or penalty enters selection; the policy allele affects survival only through the genotypes it causes to be proposed.

Each survivor first inherits its producing parent's policy. Then, independently in each survivor slot, the policy flips $M \leftrightarrow F$ with probability $\mu = 1/32$. After mutation, an F survivor has an invalid cache and an M survivor's cache becomes its producing parent's pre-update genotype. From update 2 onward every M carrier therefore holds a valid one-step lineage cache. A new $F \rightarrow M$ mutant first uses its record on the next update, and an $M \rightarrow F \rightarrow M$ history carries no older hidden record. Allele frequencies and accuracy are recorded after this transition.

2.5 Paired cells and the exact SHAM benchmark

Each of 41,600 independent blocks contains the same eight paired cells: information arm (ACTIVE or SHAM) $\times$ target law (ZERO or HALF) $\times$ policy start (ALL_F or ALL_M). Within a block, all eight cells share 32 initial uniform genotypes and all coordinate-keyed random values; caches start invalid. Random values come from counter-based Philox4x32-10 streams indexed by purpose, block, update, entity and subindex, and every draw is generated whether or not it is used, so no label, branch, arm or law changes any later random value (Supplementary Section S7).

In SHAM, the policy probe is always the fresh construction, so genotypes never depend on a label or cache. Two exact identities follow. N1: SHAM genotype, mismatch, accuracy and survivor trajectories are bit-identical between the ALL_F and ALL_M starts. N2: SHAM labels in the two starts are exact complements after every update, so their memory-use frequencies sum to 1 at every update and their start-averaged late-window frequency is exactly $1/2$ in every block. With symmetric mutation and $r = 1 - 2\mu = 15/16$, the SHAM expectations are $P(M \text{ at } t \mid \text{ALL_F}) = [1 - r^t]/2$ and $P(M \text{ at } t \mid \text{ALL_M}) = [1 + r^t]/2$. The expected ALL_M minus ALL_F difference is $r^{256} = 6.678005283720385e-08$ at update 256 and $r^{193}(1 - r^{64})/(64(1 - r)) = 9.57983820600146e-07$ averaged over the late window.

SHAM is an exact neutral benchmark for the label process (N1, N2; expected start-averaged frequency $1/2$) and a paired control for query budget and keyed random draws. Because ACTIVE genotypes and genealogies diverge from SHAM at the first valid-M retrieval, SHAM does not hold realised drift or genotype state fixed for ACTIVE paths; ACTIVE-SHAM contrasts are arm-level comparisons within the model, not path-level counterfactuals. It does not by itself separate the information carried by the cache from the mere loss of one fresh proposal; the ZERO law, under which the cache carries no lag-two information, supplies that contrast through C_{rec} . It does not represent any alternative cost, retrieval rule or memory architecture. Its neutral formulas justify the late window in expectation for SHAM only and do not establish that ACTIVE paths have converged.

2.6 Estimands, intervals and frozen decisions

For each block, let $L(\text{arm}, \text{law}, \text{start})$ be the mean memory-use frequency over updates 193-256. Define $A_HALF = [L(\text{ACTIVE}, \text{HALF}, \text{ALL_F}) + L(\text{ACTIVE}, \text{HALF}, \text{ALL_M})]/2$ and A_ZERO analogously. The four primary estimands are:

- $C_abs = A_HALF - 1/2$, the recurrent-law excess over the exact neutral expectation;
- $C_rec = A_HALF - A_ZERO$, the excess attributable to recurrence;
- $D_HALF = L(\text{ACTIVE}, \text{HALF}, \text{ALL_M}) - L(\text{ACTIVE}, \text{HALF}, \text{ALL_F})$, the start-state contrast under HALF; and
- $D_ZERO = L(\text{ACTIVE}, \text{ZERO}, \text{ALL_M}) - L(\text{ACTIVE}, \text{ZERO}, \text{ALL_F})$, the start-state contrast under ZERO.

Let $P(\text{arm}, \text{law})$ be start-averaged late-window mean population accuracy. The two secondary estimands are:

- $P_abs = P(\text{ACTIVE}, \text{HALF}) - P(\text{SHAM}, \text{HALF})$, the recurrent-law accuracy contrast; and
- $P_rec = [P(\text{ACTIVE}, \text{HALF}) - P(\text{SHAM}, \text{HALF})] - [P(\text{ACTIVE}, \text{ZERO}) - P(\text{SHAM}, \text{ZERO})]$, the recurrence interaction in accuracy.

The meaningful allele scale is $\Delta = 1/32 = 0.03125$, one expected individual in the 32-member population; it refers to a late-window expected excess, not to the persistence of one named carrier. The meaningful performance scale is $\Delta_P = 1/32$, one additional correct bit per survivor on average, which is 3.125 percentage points of accuracy.

Each family is controlled at 95% familywise coverage by two-sided Hoeffding intervals with a Bonferroni split. For a block variable with range length R and $n = 41,600$ independent blocks, the half-width is $h = R \sqrt{[\ln(2/\alpha_{\text{each}})]/(2n)}$. In the primary family, $\alpha_{\text{each}} = 0.0125$; C_abs has range length 1 and half-width 0.00781023, and C_rec , D_HALF and D_ZERO have range length 2 and half-width 0.01562046, just below $\Delta/2 = 0.015625$. In the secondary family, $\alpha_{\text{each}} = 0.025$; P_abs has range length 2 and half-width 0.01451463, and P_rec has range length 4 and half-width 0.02902925. No empirical variance, normal approximation, within-path independence, pilot or favorable pairing enters these bounds. The two families are controlled separately, not jointly.

The primary decision table was applied in order: (1) failure of any deterministic identity, seed or counter audit, saved-output reconstruction or query count gives INVALID; (2) if either D_HALF or D_ZERO has an interval not wholly inside $(-\Delta, +\Delta)$, the result is START-DEPENDENT and the question unresolved; (3) if the lower bounds of both C_abs and C_rec exceed Δ , the result SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT; (4) if the lower bound of C_abs exceeds Δ and the upper bound of C_rec is at or below Δ , enrichment is not attributable to recurrence at the fixed scale; (5) if the upper bound of C_abs is at or below Δ , the result is a bounded negative at the one-individual scale; (6) every other pattern is unresolved. Each secondary estimand was classified, in order, as meaningful positive performance if its lower bound exceeds Δ_P , meaningful adverse performance if its upper bound is below $-\Delta_P$, BOUNDED BELOW THE POSITIVE ONE-BIT SCALE if its upper bound is at or below Δ_P , and UNRESOLVED otherwise. Secondary classifications cannot alter the primary decision.

Sixteen absolute cell means, the late-window memory-use frequency and accuracy for each of the eight cells, were saved as descriptive records without inferential bounds. Together with the six inferential records they form the 22 prespecified saved estimate records. The descriptive means support interpretation only.

2.7 The start-state gate, and why allele frequency is not accuracy

The start-state gate, rule 2, asks whether populations started entirely from F and entirely from M reach the same late-window ACTIVE frequency to within one expected individual. If they did not, the late window would partly reflect where paths began rather than a late-window state approached from both starts within the frozen horizon, and no enrichment classification would be made. The gate is applied before any enrichment rule. Passing it shows only that both late-window start contrasts were bounded within $(-1/32, +1/32)$ at 95% familywise coverage over the 256-update horizon; it does not demonstrate start-independence in general, stationarity or a unique long-run state.

Allele frequency measures which policy the survivors carry; accuracy measures how well the survivors match the target. The two are linked only through the genotypes that M carriers propose. A policy can raise its carriers' representation among survivors by supplying high-scoring candidates at recurrent updates while costing a fresh proposal at every update, so its effect on mean survivor accuracy can be smaller than, or differently signed from, its effect on its own frequency. This is why the two outcomes were assigned separate estimands, scales and decision rules in advance.

3. Results

3.1 Execution, identity checks and record audit

The single production job (53530597) completed all 332,800 paths with exit code 0:0 and an elapsed time of 00:01:10. It made exactly 10,905,190,400 objective queries, 128 per update over 85,196,800 path-updates, with no N1, N2 or query-count failure. The frozen analyzer (job 53530688) and the separately implemented internal audit job (53530689) both completed with exit code 0:0. The accepted record lists 0 post-outcome changes.

Two audit tools, written separately from the producer and importing none of its code, checked the saved outputs. A C++ replay tool (MMEM-AUDIT-REPLAY-1.0.1) regenerated blocks 0-63 from block identifiers and keyed random streams and compared 64 blocks, 512 path records, 131,072 update records and 16,777,216 candidate-level audit rows against the saved outputs. After passing known-answer tests for SHA-256, Random123 Philox4x32-10 and multiply-high rejection, it reported PASS with zero mismatches. A Python verifier (MMEM-AUDIT-RECORDS-1.2.0) reconstructed all 22 saved estimates from saved path outputs, confirmed both SHAM identities in all 41,600 blocks, checked 332,800 path records and 85,196,800 update records, matched 97 manifest-listed files, recomputed the six frozen half-widths, and reported PASS with zero mismatches. These audits establish internal reproducibility of the recorded run. They were carried out within the project, partly with AI assistance; they are not external peer review and not an independent scientific replication.

3.2 The start-state gate passed

Under HALF, the late-window ACTIVE frequency from ALL_M exceeded that from ALL_F by $D_{\text{HALF}} = 0.000723$, with familywise interval $[-0.014898, 0.016343]$. Under ZERO, $D_{\text{ZERO}} = -0.000096$, with interval $[-0.015717, 0.015524]$. Both intervals lie wholly inside $(-1/32, +1/32)$; in percentage points, the two start contrasts were $+0.072$ and -0.010 . Rule 2 therefore did not apply, and the enrichment rules were evaluated.

3.3 Recurrence strongly enriched the memory-use allele

The start-averaged late-window ACTIVE memory-use frequency was 0.810292 under HALF and 0.368657 under ZERO (Table 2, Figure 2). Relative to the exact neutral expectation, $C_{abs} = 0.310292$, with familywise interval $[0.302482, 0.318102]$: under recurrence the allele exceeded the neutral 1/2 baseline by 31.029 percentage points. Relative to the nonrecurrent law, $C_{rec} = 0.441635$, with interval $[0.426015, 0.457256]$: recurrence raised the late-window frequency by 44.164 percentage points. Both lower bounds exceed $\Delta = 0.03125$ by a wide margin. Decision rule 3 applied, and the frozen primary decision is **SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT**.

The individual cell means show the same pattern from both starts. Under HALF, ACTIVE frequencies were 0.809931 from ALL_F and 0.810654 from ALL_M. Under ZERO they were 0.368705 and 0.368609, below the neutral 1/2; this ZERO deficit is descriptive, because no inferential bound on $A_{ZERO} - 1/2$ was frozen. SHAM frequencies were 0.500083 and 0.499917 under ZERO and 0.499089 and 0.500911 under HALF; each pair sums to 1, as identity N2 requires.

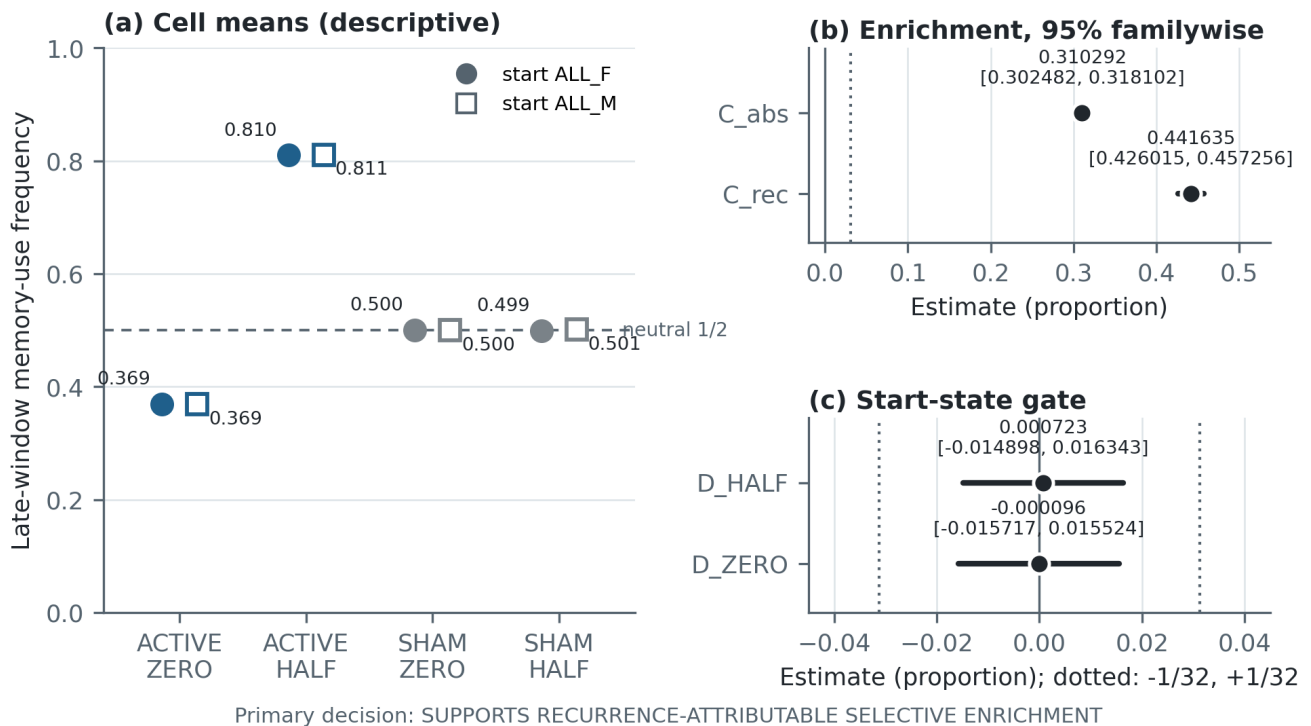


Figure 2. Allele-frequency outcomes. (a) Late-window memory-use frequency in all eight cells; markers distinguish the ALL_F and ALL_M starts, and the dashed line marks the exact neutral start-averaged expectation of 1/2. These cell means are descriptive. (b) The two enrichment estimands, C_{abs} and C_{rec} , and (c) the two start-state contrasts, D_{HALF} and D_{ZERO} , each with its frozen 95% familywise Hoeffding-Bonferroni interval; dotted lines mark the one-individual scale ($+1/32$ in b; $-1/32$ and $+1/32$ in c). Both enrichment estimands have lower bounds above $+1/32$, and both start-state contrasts lie inside the gate. All plotted values are read from the saved analyzer records at figure-generation time.

3.4 Population accuracy showed no resolved one-bit benefit

Under HALF, start-averaged ACTIVE accuracy was 0.628395, compared with 0.619018 for SHAM (Table 2, Figure 3). The recurrent-law accuracy contrast P_{abs} was 0.009377, with familywise interval $[-0.005138, 0.023892]$: an estimated contrast of +0.938 percentage points whose upper bound, 2.389 points, lies below the prespecified 3.125-point one-correct-bit threshold. The frozen classification is **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**. Because this interval also includes zero, even the sign of the recurrent-law accuracy contrast is not resolved at the familywise level.

The recurrence interaction P_{rec} was 0.011360, with interval $[-0.017669, 0.040390]$, or +1.136 percentage points with an interval from -1.767 to +4.039 points. The interval spans both zero and the one-bit scale, and the frozen classification is **UNRESOLVED**. The P_{rec} half-width, 0.02902925, is only slightly below Δ_P , so this design had limited capacity to resolve the interaction at the one-bit scale; the half-width was fixed in the frozen specification.

Under ZERO, start-averaged ACTIVE accuracy was 0.605327, compared with 0.607310 for SHAM: a small adverse descriptive contrast of -0.001983, or -0.198 percentage points. It carries no inferential bound and is retained as reported. SHAM accuracies were identical between starts (0.607310 under ZERO and 0.619018 under HALF), as identity N1 requires. The analyzer recorded the descriptive relation of P_{abs} to allele enrichment as DISAGREEMENT and that of P_{rec} as UNCERTAINTY.

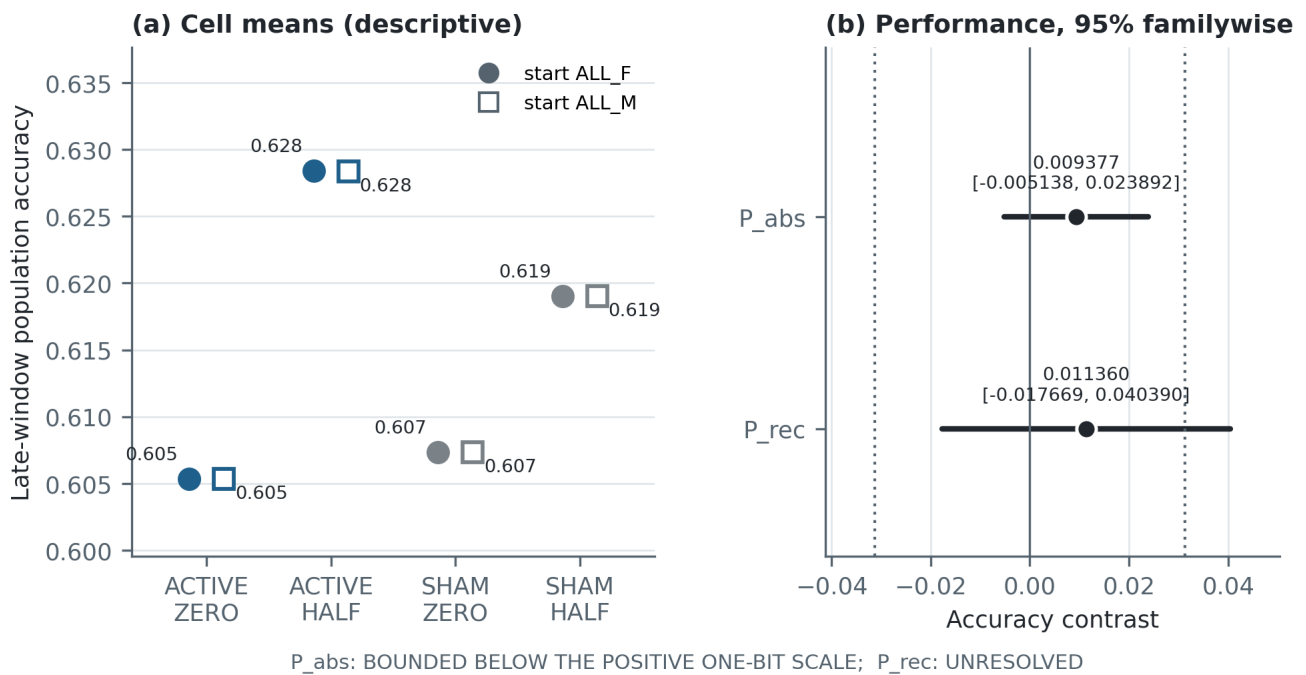


Figure 3. Population-accuracy outcomes. (a) Late-window mean population accuracy in all eight cells (descriptive); SHAM accuracies coincide between starts by identity N1. (b) The two secondary estimands with their frozen 95% familywise Hoeffding-Bonferroni intervals; dotted lines mark the one-correct-bit scale ($-1/32$ and $+1/32$). P_{abs} is bounded below the positive one-bit scale and P_{rec} is unresolved. All plotted values are read from the saved analyzer records at figure-generation time.

Table 1. Inferential estimands and frozen classifications. Intervals are two-sided 95% familywise Hoeffding-Bonferroni intervals over 41,600 blocks, computed separately for the primary and secondary families.

Estima nd	Estimate	95% familywise interval	Half-width	Frozen classification
C_abs	0.31029 2	[0.302482, 0.318102]	0.007810 23	Lower bound exceeds 1/32
C_rec	0.44163 5	[0.426015, 0.457256]	0.015620 46	Lower bound exceeds 1/32
D_HALF	0.00072 3	[-0.014898, 0.016343]	0.015620 46	Inside (-1/32, +1/32)
D_ZERO	-0.00009 6	[-0.015717, 0.015524]	0.015620 46	Inside (-1/32, +1/32)
P_abs	0.00937 7	[-0.005138, 0.023892]	0.014514 63	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE
P_rec	0.01136 0	[-0.017669, 0.040390]	0.029029 25	UNRESOLVED

Table 2. Descriptive late-window cell means (no inferential bounds were frozen for these 16 records).

Cell	Memory-use frequency	Population accuracy
ACTIVE, ZERO, ALL_F	0.368705	0.605324
ACTIVE, ZERO, ALL_M	0.368609	0.605330
ACTIVE, HALF, ALL_F	0.809931	0.628405
ACTIVE, HALF, ALL_M	0.810654	0.628385
SHAM, ZERO, ALL_F	0.500083	0.607310
SHAM, ZERO, ALL_M	0.499917	0.607310
SHAM, HALF, ALL_F	0.499089	0.619018
SHAM, HALF, ALL_M	0.500911	0.619018

3.5 The joint result

Taken together, the primary and secondary families give one result rather than two. Lag-two recurrence strongly enriched a mutation-generated memory-use allele despite its fixed cost of one fresh proposal, both late-window start contrasts passed the prespecified finite-horizon $\pm 1/32$ start-state gate, and the enrichment was attributable to recurrence at the one-individual scale. Over the same paths, the recurrent-law ACTIVE-SHAM accuracy contrast was bounded below one correct bit per survivor with sign unresolved, and whether recurrence changed the accuracy effect of the allele was unresolved. In this model, selection on the use of a supplied cache was strong, while the ACTIVE-SHAM accuracy contrast was small at most and was not resolved at the prespecified scale.

4. Discussion

The central observation is a separation between what selection favored and what the population gained. Under the aligned recurrent law, more than four-fifths of late-window survivors carried the memory-use allele in both start conditions over the 256-update horizon, whereas the ACTIVE-SHAM accuracy contrast, in which SHAM labels are inert, was bounded below one additional correct bit per survivor. Under the nonrecurrent law, where the cache carried no lag-two information, the allele fell descriptively below its neutral frequency, and ACTIVE accuracy was descriptively slightly below SHAM.

The study was not designed to identify the mechanism of this separation, and the following explanations are hypotheses for future work rather than findings. Survival depends on the relative scores of candidates within an update. A carrier whose cache matches a recurrent target supplies a candidate that outscores most fresh proposals, so carriers can be strongly overrepresented among survivors even if the survivor set, which is dominated by the best candidates from all families, improves only modestly on average. At non-recurrent updates the same carriers have given up one fresh draw, and the local child of every parent already supplies a near-copy of the best current candidate, which may limit what a retrieved record adds to the population's mean fit. The ZERO accuracy contrast is in the direction expected from displaced exploration, but it is small and descriptive.

These results are consistent in spirit with prior findings that the marginal benefit of learning or memory can be small even where memory is favored [4], and with models in which memory-related loci or anticipatory effects evolve only under particular environmental regimes [5, 7]. The present design adds an exact neutral benchmark for a mutable use allele, a cost expressed in the same currency as search itself, and prospectively separate scales for enrichment and performance. Its main methodological implication is that an allele-frequency response to environmental structure, however strong, should not be read as a population-performance response without a separately stated performance estimand and scale.

The result is deliberately narrow. HALF matches the recurrence lag to the retrieval delay and so favors retrieval by construction; no other lag or copy probability was tested. The opportunity cost is specific to replacing one effectively uniform proposal in a four-candidate budget. The cache architecture was supplied rather than evolved. Within those conditions, the primary inference is strong and the secondary inference is bounded or unresolved; neither should be generalized beyond them.

5. Limitations

- **One aligned recurrent law.** HALF was a prior, outcome-informed choice that matches the cache delay by construction. No other lag, copy probability or target law was tested, and the result does not establish selection under arbitrary recurrence.
- **One fixed cost.** The cost is the replacement of one of two effectively uniform fresh proposals. It is a model-specific search-budget trade-off, not a calibrated energetic or biological cost and not a general cost scale. No other cost was tested.
- **Supplied architecture.** The one-record private cache and its retrieval rule were supplied. Only the use policy mutated. The study does not address the origin of memory architecture or capacity.
- **Retention and retrieval are not separated.** The allele jointly determines whether the one-step record is retained and whether it is retrieved; every individual has a cache slot, but only M carriers receive a valid record, so this design cannot separate selection on retention

from selection on retrieval.

- **Finite horizon; start-state gate only.** Both late-window start contrasts were bounded within one expected individual (95% familywise) over 256 updates. This is not a demonstration of general start-independence, stationarity, equilibrium, evolutionary stability, rare-mutant invasion or fixation probability.
- **Performance not resolved at the one-bit scale.** P_{abs} excludes a gain of one correct bit but does not resolve its sign; P_{rec} is unresolved, and its half-width is close to the one-bit scale. Population benefit was not demonstrated. The small adverse ZERO accuracy contrast and the ZERO allele deficit are descriptive.
- **Separate families.** The primary and secondary families each have 95% familywise coverage; coverage is not controlled jointly across both families or across the 16 descriptive means.
- **What SHAM does not control.** SHAM is an exact neutral benchmark for label dynamics, not a control for alternative caches, alternative costs or alternative retrieval rules. ACTIVE and SHAM genotypes and genealogies diverge at the first valid-M retrieval, so ACTIVE-SHAM contrasts are arm-level comparisons, not path-level counterfactuals.
- **Abstract model.** The genotype, objective, selection rule and population are abstract. No biological, ecological, neural or engineering extrapolation is supported, and the study does not replicate earlier internal private-memory studies.
- **Internal audit only.** Replay and record verification were performed by separately written tools within the project, partly with AI assistance. They are not external peer review or an independent scientific replication. Producer and audit tools were both written with AI assistance from the same specification; the audits detect implementation divergence, not a shared misreading of the specification.
- **Bounded literature search.** The literature comparison rests on a structured search of open scholarly metadata with targeted citation chaining, not an exhaustive systematic review; it supports a bounded distinct-contribution claim only and cannot establish a unique nearest precedent. The search archive will be included in the public package (Section 7).

6. Methods summary

The model, cohorts and analysis follow the frozen specification PHASE2-MUTABLE-MEMORY-001 revision 1 exactly; Supplementary Sections S2-S10 give the full update order, target laws, allele mutation, cache inheritance and use, candidate budget, selection, estimands, interval construction and decision rules. Random values were drawn from canonical Random123 Philox4x32-10 streams whose keys are derived from SHA-256 of a purpose-specific namespace string and whose counters are the coordinates (block, update, entity, subindex); a coordinate-schema enumeration and collision audit preceded production. Ten deterministic fixtures, including hand traces, memory-better and memory-worse cases, duplicate and invalid-cache cases, full SHAM identity paths, target-law pairing, an exact weighted-selection fixture with a forced rejection, random-coordinate invariance and single-path regeneration, passed before production; none contains a sampled scientific outcome.

Per-update binary records were saved for every path, per-path late-window summaries and final-state hashes for every path, and full candidate-level rows for the fixed audit blocks 0-63. The frozen analyzer computed the 22 estimate records, their bounds and classifications from saved path outputs. Interval endpoints use exact rational estimates and high-precision half-widths. Descriptive diagnostics (mutation counts, cache-probe use and survival, and occupancy counters) were saved for interpretation only (Supplementary Section S12).

This is an abstract computational model. It involves no human participants, no animals and no personal data.

7. Data and code availability

A [public reproducibility package](#) accompanies this preprint. It contains the frozen specification, source code, configuration, build environment and SHA-256 manifests; all per-path and per-block outputs; analyzer records; internal audit receipts; the prospective design-review, production-authorization, production-submission and acceptance records; the literature-search archive (search script, API responses, deduplicated records and title screen); and this manuscript source with its figure and statistics-checking scripts. The complete per-update outputs and candidate-level audit rows are distributed as a separate complete-outcomes archive linked from the repository release. Manuscript text, figures and data are released under the Creative Commons Attribution 4.0 International license (CC BY 4.0), and original code under the MIT License. The authenticated outputs used here have the following SHA-256 values: decision record 5b3aea3c18c4530b2be9f29d5c64fdd8173b6b323823bf9cc3361bd4e7010fc0; 22 estimate records 83828a49936bb8364813e2634f11cb28c4de2212b4b162d96e616ddbc7b4f8c0; descriptive diagnostics 51554620a5e302416edf8de8a073c6962abbc2c8967da80b91b83e4c3d5b685c; replay receipt 9b40322d0d09b90f0a6862533c1efdc7aff21bd699a882b9b9555c5f7c4b4857; record-verification receipt afd96dc0a5b190401c035fc66c8af8fd4d4603e7185d05bff73fb05e698cff5a; production inventory audit 622fa9e01a40c44ba545926e54b06e102d20b7fcf9e9c8e645ddaf9f7212e09c.

Prospective timing is recorded as follows (all times UTC, 2026-10-01). The frozen specification, design/PHASE2-MUTABLE-MEMORY-001-REV1.md (revision 1), has SHA-256 c17a3a1e9ac9cf2260f6743eaf08e4c2808080a767af16d2f9e9d2b9ad3e821d. Its terminal design review by a non-implementing AI reviewer (record CLAUDE-003, verdict GO) was completed at 01:55:25; the review has SHA-256 787cf32a20312f2682c761243f3dec04c5eb8036d14d6efc444e9830db37960a and its receipt 6277757f9ab3dfa611515c161776c17b561738eba6fa478ae0f8634e13c03176. The production execution authority (SHA-256 ee3cb68bed5d212e6ea10ece542e2e90c32cb9cea07d85fceda10fc56b375b1d) was issued at 12:53:04, and the sole production job (53530597) was submitted at 12:55:19 with no automatic retry and no parameter changes. The acceptance record has SHA-256 2eb3aaf5d754f4570a75522aa9e1f38358715c06c674d5a267732a1357926e6f. The frozen design and its GO review therefore precede the only production authorization and submission.

8. AI assistance

AI systems assisted with design criticism, implementation, code review, analysis checking, literature organization and drafting, under human direction. The author directed the work, made the scientific decisions and is responsible for the content. Internal reviews performed by AI models, including the non-implementing AI review of the frozen specification, are not external peer review.

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